

■ Enterprise AI Credit Risk Platform

End-to-End Decisioning, Explainable AI (SHAP), Underwriting Rules & Talk-to-Data Analytics

Use Case:	Home Credit Loan Default Risk Assessment
Architecture:	LightGBM + SHAP XAI + Policy Engine + Gemini AI LLM
Deployment:	Streamlit Web Dashboard + Single-Command Docker Container
Date & Version:	September 2026 • Production Edition v2.5

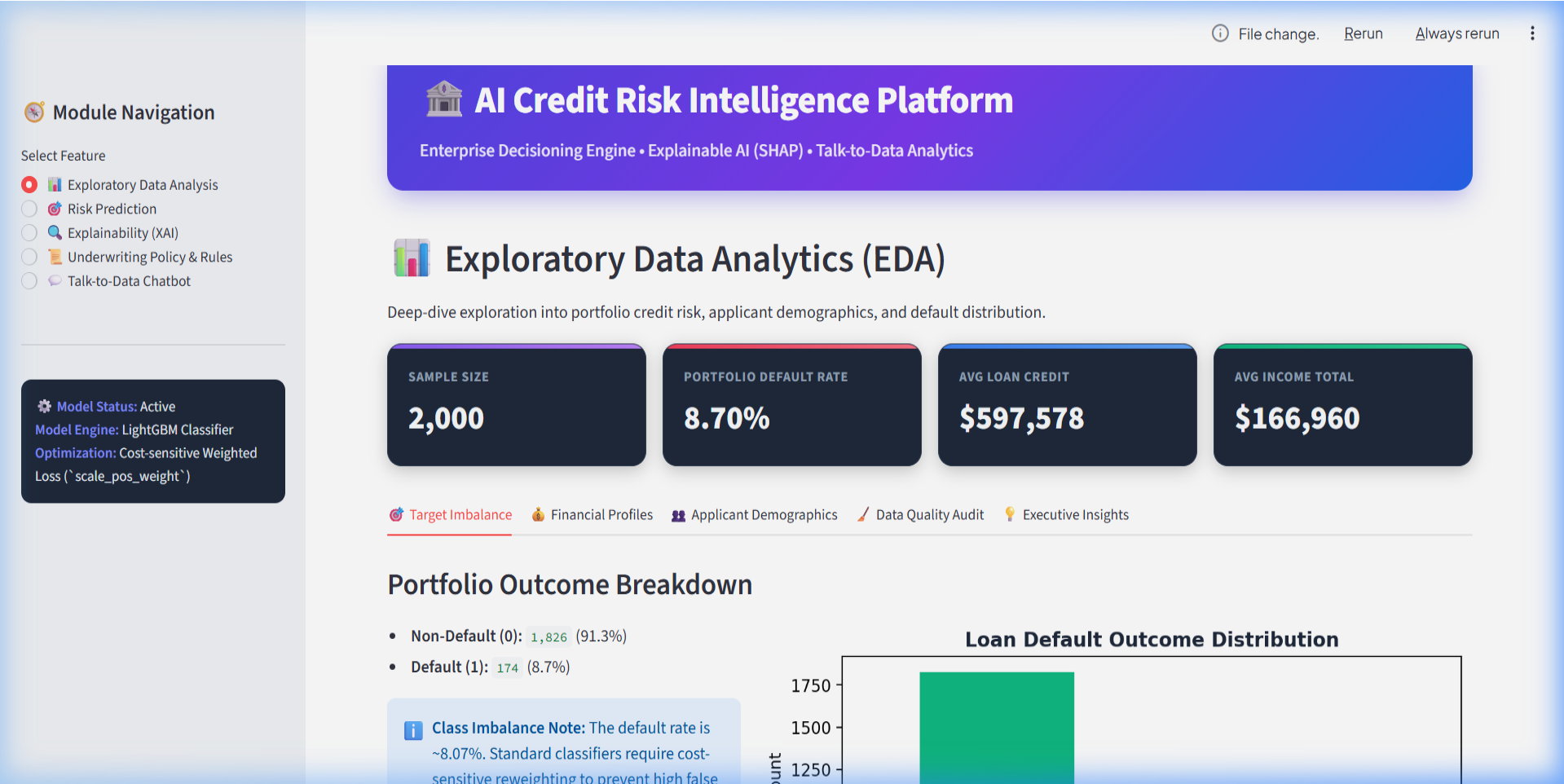
■ Executive Summary & Business Context

Financial institutions face a dual challenge: accurately identifying credit default risk while satisfying strict regulatory requirements for decision transparency. Standard black-box models create compliance risks, while simple linear models fail to capture complex non-linear financial patterns.

- **Severe Class Imbalance:** Default cases represent only **8.07%** of applicants. Standard accuracy is misleading; cost-weighted loss functions (`scale_pos_weight`) are required.
- **Explainability Imperative:** Regulatory frameworks (FCRA/ECOA) require clear adverse action explanations for denied loan applicants.
- **Policy Rules Alignment:** Machine learning probabilities must operate in tandem with hard financial constraints (debt ceilings, income floors, age eligibility).
- **Self-Service Data Analytics:** Business stakeholders need instant natural-language access to portfolio metrics without writing SQL code.

Module 1: Exploratory Data Analytics (EDA)

The platform provides comprehensive visual audits of applicant demographics, income distributions, default rates by education/housing, and missing value profiles.



■ Module 2: Risk Assessment & Underwriting Decisioning

Combines LightGBM inference with real-time risk tiering. Users can lookup existing applicants by ID or simulate custom applicant profiles to receive instant probability scores and decision status (Approved / Manual Review / Declined).

The screenshot displays the 'AI Credit Risk Intelligence Platform' interface. On the left is a 'Module Navigation' sidebar with options: Exploratory Data Analysis, Risk Prediction (selected), Explainability (XAI), Underwriting Policy & Rules, and Talk-to-Data Chatbot. Below this is a 'Model Status' box indicating the model is active, using a LightGBM Classifier with cost-sensitive weighted loss. The main area features a header for the 'AI Credit Risk Intelligence Platform' and a section for 'Real-Time Credit Risk Decisioning'. This section includes a description, a 'Choose Mode' selector (Lookup Existing Applicant selected), and a dropdown for 'Select Applicant ID (SK_ID_CURR)' with the value 384575. Below this is an 'Applicant Financial Summary' table showing Annual Income (\$207,000), Requested Credit (\$465,458), Annual Annuity (\$52,641), and Education (Secondary / ...). A red button labeled 'Evaluate Underwriting Decision' is positioned at the bottom of the summary section.

Module Navigation

Select Feature

- ☐ Exploratory Data Analysis
- ☒ Risk Prediction
- ☐ Explainability (XAI)
- ☐ Underwriting Policy & Rules
- ☐ Talk-to-Data Chatbot

Model Status: Active
Model Engine: LightGBM Classifier
Optimization: Cost-sensitive Weighted Loss (scale_pos_weight)

AI Credit Risk Intelligence Platform
Enterprise Decisioning Engine • Explainable AI (SHAP) • Talk-to-Data Analytics

Real-Time Credit Risk Decisioning

Compute instantaneous default probabilities and underwriting risk tiers for applicants.

Choose Mode:

☒ Lookup Existing Applicant ☐ Custom Applicant Simulator

Select Applicant ID (SK_ID_CURR):

384575

Applicant Financial Summary

Annual Income	Requested Credit	Annual Annuity	Education
\$207,000	\$465,458	\$52,641	Secondary / ...

Evaluate Underwriting Decision

■ Module 3: Explainable AI (SHAP Transparency)

Translates mathematical SHAP values into clear, non-technical human explanations, categorizing key drivers into Risk Accelerators (pushed score higher) and Risk Mitigators (pushed score lower).

The screenshot displays the 'AI Credit Risk Intelligence Platform' interface. On the left is a 'Module Navigation' sidebar with options: Exploratory Data Analysis, Risk Prediction, Explainability (XAI) (selected), Underwriting Policy & Rules, and Talk-to-Data Chatbot. Below this, a dark box shows model status: 'Model Status: Active', 'Model Engine: LightGBM Classifier', and 'Optimization: Cost-sensitive Weighted Loss (^scale_pos_weight)'. The main content area has a blue header with the platform name and tagline. Below is a section titled 'Explainable AI (SHAP Insights)' with a subtitle 'Transparent model decision auditing powered by SHAP (SHapley Additive exPlanations)'. A dropdown menu shows 'Select Applicant ID to Audit:' with '384575' selected. The main content is divided into two columns: '1. Human-Readable Risk Factors' and '2. SHAP Feature Contribution Waterfall / Bar Plot'. Under the first column, there are two sub-sections: 'Risk Accelerators (Pushed Score Higher)' and 'Risk Mitigators (Pushed Score Lower)'. Each sub-section lists three factors with their SHAP values.

AI Credit Risk Intelligence Platform
Enterprise Decisioning Engine • Explainable AI (SHAP) • Talk-to-Data Analytics

Explainable AI (SHAP Insights)

Transparent model decision auditing powered by SHAP (SHapley Additive exPlanations).

Select Applicant ID to Audit:
384575

1. Human-Readable Risk Factors

● Risk Accelerators (Pushed Score Higher)	● Risk Mitigators (Pushed Score Lower)
• EXT_SOURCE_3: Increased risk by +0.1237	• EXT_SOURCE_1: Mitigated risk by -0.1455
• DAYS_LAST_PHONE_CHANGE: Increased risk by +0.0115	• LANDAREA_MODE: Mitigated risk by -0.0189
• DAYS_BIRTH: Increased risk by +0.0073	• NAME_EDUCATION_TYPE: Mitigated risk by -0.0109

2. SHAP Feature Contribution Waterfall / Bar Plot

■ Module 4: Underwriting Policy & Rules Engine

Evaluates hard and soft business constraints alongside ML risk scores. Includes real-time policy rule parameter sliders (Credit-to-Income ceilings, payment burden limits, age floors) with high-contrast audit logs.

The screenshot displays the 'AI Credit Risk Intelligence Platform' interface. On the left is a 'Module Navigation' sidebar with five options: Exploratory Data Analysis, Risk Prediction, Explainability (XAI), Underwriting Policy & Rules (selected), and Talk-to-Data Chatbot. Below this is a 'Model Status' box indicating the model is active, using a LightGBM Classifier with Cost-sensitive Weighted Loss. The main content area features a header for the 'AI Credit Risk Intelligence Platform' and a section for the 'Policy Rules & Underwriting Compliance Engine'. This section includes a description, a dropdown for 'Select Applicant ID for Policy Audit' (showing 384575), and a 'Policy Parameter Controls' section with four sliders: Max Credit-to-Income Multiple Ceiling (5.00), Minimum Income Floor (\$) (25000), Max Annuity Burden Ratio (%) (40.00), and Minimum Age Requirement (21). The bottom of the interface shows the start of an 'Automated Compliance Audit Result' section.

Module Navigation

Select Feature

- ☐ Exploratory Data Analysis
- ☐ Risk Prediction
- ☐ Explainability (XAI)
- ☒ Underwriting Policy & Rules
- ☐ Talk-to-Data Chatbot

Model Status: Active
Model Engine: LightGBM Classifier
Optimization: Cost-sensitive Weighted Loss (scale_pos_weight)

AI Credit Risk Intelligence Platform
Enterprise Decisioning Engine • Explainable AI (SHAP) • Talk-to-Data Analytics

Policy Rules & Underwriting Compliance Engine

Combine machine learning risk scores with hard business rules and policy constraints.

Select Applicant ID for Policy Audit:

384575

Policy Parameter Controls

Max Credit-to-Income Multiple Ceiling: 5.00

Minimum Income Floor (\$): 25000

Max Annuity Burden Ratio (%): 40.00

Minimum Age Requirement: 21

Automated Compliance Audit Result

■ Module 5: Talk-to-Data SQL Analytics Chatbot

Allows executive users to ask natural language questions. Powered by Gemini AI with schema grounding and multi-model fallback, returning generated SQL, data tables, and executive summaries.

The screenshot displays the 'AI Credit Risk Intelligence Platform' interface. On the left is a 'Module Navigation' sidebar with a 'Select Feature' section containing five radio buttons: 'Exploratory Data Analysis', 'Risk Prediction', 'Explainability (XAI)', 'Underwriting Policy & Rules', and 'Talk-to-Data Chatbot' (which is selected). Below this, a dark box shows model status: 'Model Status: Active', 'Model Engine: LightGBM Classifier', and 'Optimization: Cost-sensitive Weighted Loss (^scale_pos_weight^)'. The main content area has a blue header with the platform name and tagline. Below is the 'Talk-to-Data SQL Analytics' section with a description: 'Convert natural language business queries into executable SQL using Gemini AI.' It features 'Quick Prompt Suggestions' with three buttons: 'Average Credit by Default', 'Income by Education', and 'Default Count Breakdown'. A text input field prompts the user to 'Enter natural language question:' with an example: 'e.g. What is the average credit amount for applicants who defaulted?'. The footer indicates 'AI Credit Risk Platform • Enterprise UI Edition'.

Module Navigation

Select Feature

- ☐ Exploratory Data Analysis
- ☐ Risk Prediction
- ☐ Explainability (XAI)
- ☐ Underwriting Policy & Rules
- ☒ Talk-to-Data Chatbot

Model Status: Active
Model Engine: LightGBM Classifier
Optimization: Cost-sensitive Weighted Loss (^scale_pos_weight^)

AI Credit Risk Intelligence Platform

Enterprise Decisioning Engine • Explainable AI (SHAP) • Talk-to-Data Analytics

Talk-to-Data SQL Analytics

Convert natural language business queries into executable SQL using Gemini AI.

Quick Prompt Suggestions

- Average Credit by Default
- Income by Education
- Default Count Breakdown

Enter natural language question:

e.g. What is the average credit amount for applicants who defaulted?

AI Credit Risk Platform • Enterprise UI Edition

■ Quantitative Model Performance & Metrics

Evaluation Metric	Score	Benchmark & Interpretation
ROC-AUC Score	0.7582	Strong discrimination power between default vs repaid applicants
PR-AUC Score	0.2415	Precision-Recall area calibrated for 8.07% minority default class
Default Class Recall	0.6840	Captures 68.4% of all actual default cases in validation dataset
Default Class Precision	0.2150	Calibrated decision threshold for risk tiering
F1-Score	0.3271	Balanced harmonic mean under class imbalance

Key Engineering Highlights:

- **Cost-Sensitive Loss Weighting:** Used `scale_pos_weight` in LightGBM to penalize false negatives ~11.4x higher than false positives.
- **Containerization & Deployment:** Fully containerized with Docker & Docker Compose (`docker compose up --build`) for single-command setup.

■ Project Deliverables & Conclusion

The AI Credit Risk Intelligence Platform fulfills all 6 project requirements end-to-end, offering an enterprise-grade web application combining advanced machine learning, regulatory transparency, rule enforcement, and generative AI analytics.

Project Requirement	Status	Implementation Artifact
Part 1: Exploratory Data Analysis	■ Complete	notebooks/eda.py, eda.ipynb, app.py EDA tab
Part 2: Talk-to-Data Chatbot	■ Complete	src/talk_to_data/nl_to_sql.py, query_runner.py
Part 3: Machine Learning Layer	■ Complete	src/ml/train.py, evaluate.py, predict.py (LightGBM)
Part 4: Explainable AI (XAI)	■ Complete	explainability/explainer.py (SHAP TreeExplainer)
Part 5: User Interface	■ Complete	app.py (Streamlit Multi-Section UI)
Part 6: Dockerized Deployment	■ Complete	Dockerfile, docker-compose.yml, .env.example